The impact of emerging technologies

Every emerging technology changes the world it lands in — and almost never in just one direction. The same system that lifts one person up can leave another behind. This module is about naming those impacts precisely, the way the exam wants you to.

What this key knowledge covers
This summary distils everything you need for **KK03 — The impact of emerging technologies** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Meet the scenario: Bellview & the Tran family

Bellview is a regional Victorian town built around one big employer — *ParcelHub*, a parcel distribution warehouse. This year ParcelHub switched on an automated sorting line, a fleet of warehouse “pick-bots,” and an app-only returns kiosk. We’ll follow the ripple through one family:

Spent 12 years on the ParcelHub floor reading labels and sorting parcels. His job is the front line of **automation**, **deskilling** and **job loss**.

Banking is now app-only and the shops are self-checkout. She meets the edges of **access** and the **decline of human interaction**.

An anonymous “school confessions” app and endless group chats expose her to **cyberbullying** and a quiet drop in face-to-face **interpersonal skills**.

Sees soaring **productivity**, cheaper delivery for the whole town, a **sustainability** win on truck idling — and a **misuse**/e-waste headache.

One town, one family — but the impacts land at three levels: the **individual**, the **organisation**, and **society**. Keep that ladder in mind; the exam loves it.

One technology, three levels of impact

Drop an emerging technology into the centre and watch the impact **ripple outward** — first to the individual, then the organisation, then society. Tap a technology to switch the source. Notice that no impact stops at one ring.

Impact of current & emerging technologies

This strand is about how technology reshapes **people and relationships** — how we work, communicate and treat each other. Three named impacts you must be able to explain:

automation Automation

Automation means a task is completed **without human intervention** — a machine, robot or program does what a person used to do. At ParcelHub, pick-bots now move parcels Đức once carried by hand.

cyberbullying Cyberbullying

Cyberbullying is using a digital system to **threaten, harass, intimidate or humiliate** someone. Emerging tech makes it worse along three axes: it's *scale* (one post reaches the whole year level), *persistence* (it's saved, screenshotted, resurfaced), and *anonymity* (Bellview High's "confessions" app hides who posted).

human interaction Decline of physical interaction & interpersonal skills

When self-checkouts, app banking and messaging replace face-to-face contact, two things slip: the **amount** of real human interaction, and the **skill** of doing it well — reading body language, small talk, resolving conflict in person. Bà used to chat with the bank teller every week; now she taps an app and speaks to no one. Mai's group can text for hours but freezes in a face-to-face disagreement.

Economic issues involving emerging technologies

This strand zooms out to **money, work and resources** — who can afford the tech, what happens to jobs and skills, and what it costs the planet. Five named issues:

Who can **get** and **afford** the technology? New tools can widen the *digital divide* — Bà can't use app-only banking, a low-income family can't afford the device the school assumes everyone has.

When machines take over the skilled part of a job, the human is left doing a **narrower, simpler** task — and the original expertise fades. Đức went from reading and routing parcels to just clearing jams the bot can't.

Roles disappear entirely as automation scales. ParcelHub cut a third of its floor staff. New jobs appear too (bot technicians) — but rarely for the **same people** in the **same town**.

Technology used for harm or beyond its intended purpose — scams using AI voices, surveillance creep, selling data collected for one purpose to another. ParcelHub's customer data could be on-sold without consent.

The environmental and long-term economic footprint. **Both directions again:** ParcelHub's smart routing cuts truck idling and fuel (a win) — but the constant device upgrades and dumped scanners create **e-waste**, and data centres burn real energy (a cost).

The same technology, weighed from two sides

Here's the idea that lifts answers into the top band, made physical. Pick a technology, then choose **whose** eyes you're looking through. The prism splits the technology into benefit-rays and cost-rays, and the scale tips toward whichever side weighs more **for that stakeholder**.

Switch the lens and watch it re-balance — the technology didn't change, the stakeholder did.

Watch deskilling and job loss happen

Drag the automation level on ParcelHub's sorting line from **manual** to **fully automated**. As bots take over each station, three readouts move together: productivity climbs, headcount falls (job loss), and the workers who remain do narrower tasks (deskilling). This is the economic strand in one motion.

Flip each impact to its other edge

Front = a familiar framing. Tap to flip to the **edge people forget**. Training your brain to always reach for the second edge is exactly the habit the exam rewards.

faster, cheaper, 24/7 output

…but the same automation causes deskilling and job loss for the workers it replaces.

…but creates an access barrier for people like Bà and removes everyday human interaction.

…but anonymity, scale and persistence make them engines for cyberbullying.

…but enables misuse (scams, deepfakes) and carries an energy + e-waste cost.

…but constant device upgrades dump old hardware as e-waste — the other side of sustainability.

…but for many it erodes face-to-face contact and interpersonal skills.

Social impact or economic issue?

Drag (or tap) each impact from Bellview into the right bucket. Mixing the two strands is one of the most common ways students lose easy marks.

Do exactly what the verb asks

The mark is for matching your answer to the command word. Tap one to see what it demands and how it changes a KK03 answer.

Build a top-band paragraph live

A reliable structure for "explain/analyse/discuss the impact" questions: **P**oint (name the impact + stakeholder) → **E**vidence (a concrete example, ideally from the scenario) → **L**ink (back to the question + the other edge). Write below — the meters react to what you include.

Five questions · 18 marks · self-marked

Each question reveals a deliberately weak answer (with why it drops marks), a strong answer, and the **marking guide**. Mark yourself honestly — your score fills the dock in the corner.

Common traps & misconceptions

01 Writing “good” or “bad,” full stop
“Automation is bad because it takes jobs.” One edge only — and no stakeholder. ✓ **Fix:** Show **both edges** and name **who** gains vs loses: a productivity benefit for ParcelHub, a job-loss cost for Đức.

02 Confusing deskilling with job loss
Calling Đức’s shrunken role “job loss” when he still has a job, or vice-versa. ✓ **Fix: Deskilling** = keeps the job, narrower. **Job loss** = role removed. Use the exact term the case fits.

03 Mislabelling the strand
Listing *access* as a “social impact,” or *cyberbullying* as an “economic issue.” ✓ **Fix:** Social strand = automation, cyberbullying, decline of interaction. Economic = access, deskilling, job loss, misuse, sustainability.

04 Vague, scenario-free examples
“Robots take jobs and people get sad.” No named stakeholder, no concrete instance. ✓ **Fix:** Anchor it: “ParcelHub’s pick-bots made a third of floor staff redundant.” Specifics earn the mark.

05 Treating cyberbullying as “not a tech impact”
Saying bullying always existed, so it isn’t about technology. ✓ **Fix:** Name the **amplifying** role: scale, persistence and anonymity that emerging apps add.

06 Ignoring the command word
Listing impacts when asked to *analyse*, *discuss* or *evaluate*. ✓ **Fix:** Match the verb: analyse = examine cause/effect; discuss = both sides; evaluate = weigh + judge.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define · 2 marks

At ParcelHub, pick-bots now move parcels that staff once carried by hand. Define **automation** and give one example of it from the ParcelHub workplace. ()

✗ A weak answer

— why it loses marks “Automation is when computers and robots do things instead of people, like AI.” Misses the key phrase **without human intervention**, and the example is generic (“AI”), not from the workplace given.

✓ A strong answer

Definition: Automation is a process or task completed **without human intervention** — a machine or program does what a person previously did. **Example:** ParcelHub’s pick-bots move and sort parcels automatically, a task previously done by hand by floor staff.

How the marks are awarded

- 1 mark — definition includes the idea of a task done **without human intervention**.
- 1 mark — a relevant example drawn from the ParcelHub workplace (e.g. pick-bots sorting parcels).

Q2. Distinguish · 3 marks

After automation, Đức no longer routes parcels; he only clears occasional jams. Meanwhile a third of the floor staff were made redundant. **Distinguish** between **deskilling** and **job loss**, referring to ParcelHub. ()

✗ A weak answer

— why it loses marks “Deskilling and job loss both mean automation takes people’s jobs at ParcelHub because robots are better than humans.” Treats the two terms as the same thing. The command word is **distinguish** — the answer must make the difference explicit, which this doesn’t.

✓ A strong answer

Deskilling is when automation takes over the skilled part of a role, so the worker keeps a job but it becomes **narrower and simpler** — e.g. Đức now only clears jams instead of reading and routing parcels. **Job loss** is when the role is **removed entirely** — e.g. a third of ParcelHub’s floor staff were made redundant. **Key difference:** in deskilling the person is still employed (with reduced skills used); in job loss the position no longer exists.

How the marks are awarded

- 1 mark — correct meaning of deskilling, ideally tied to Đức.
- 1 mark — correct meaning of job loss, ideally tied to the redundancies.
- 1 mark — an explicit statement of the **difference** between them.

Q3. Analyse · 4 marks

Bellview’s bank and several shops are now app-only or self-service. This suits busy commuters but Mai’s grandmother, Bà, struggles with it. **Analyse** the impact of moving to app-only services on **two different** members of the Bellview community. ()

✗ A weak answer

— why it loses marks “App services are good because they’re fast and convenient and everyone can use their phone to do things quickly.” One-sided and one-stakeholder. To **analyse**, you must examine effects on **different** people and show cause and effect — not just list a benefit.

✓ A strong answer

Connected commuter (benefit): app-only services add convenience and speed — banking and shopping in seconds, anytime, without travelling. The technology **improves access** for those already digitally able. **Bà (cost):** the same shift creates an **access barrier** — she finds the app hard to use — and removes the **human interaction** she used to have with the teller, contributing to a decline in everyday face-to-face contact. **Analysis:** the identical technology produces opposite effects depending on the user’s digital confidence and need for contact — a clear example of **impact being double-edged**.

How the marks are awarded

- 1 mark each — identifies an impact on two **different** stakeholders (max 2).
- 1 mark — explains a **benefit** with cause/effect.
- 1 mark — explains a **cost** (access barrier and/or decline in interaction) with cause/effect.

Q4. Discuss · 4 marks

A different school, Riverside College, introduces “StudyBuddy,” a generative-AI tutoring app that writes feedback and practice questions for every student. **Discuss** the impact of StudyBuddy, including **one social** and **one economic** impact, and at least one benefit and one cost. ()

✗ A weak answer

— why it loses marks “StudyBuddy is helpful because AI is smart and gives students answers, so they will get better marks and learn more easily.” Only benefits, only one dimension. **Discuss** requires both sides, and the question explicitly asks for a **social** and an **economic** impact — neither is clearly labelled here.

✓ A strong answer

Social (benefit): instant, personalised feedback can support students who can’t afford a private tutor, and help those who are shy about asking the teacher. **Social (cost):** over-reliance may reduce **interpersonal learning** — fewer questions asked of teachers and peers — and AI can produce **biased or incorrect** feedback if unverified. **Economic (access):** StudyBuddy only helps students with a device and connection, so it can **widen the digital divide** for low-income families; there’s also a **sustainability/energy** cost in running large AI models. **Overall:** the benefit (access to personalised help) is real but uneven, so the impact depends on which students you ask.

How the marks are awarded

- 1 mark — a clearly labelled **social** impact.
- 1 mark — a clearly labelled **economic** impact (e.g. access, sustainability).
- 1 mark — at least one **benefit** and one **cost** are both present.
- 1 mark — a concluding comment that **weighs** the sides (discussion, not a list).